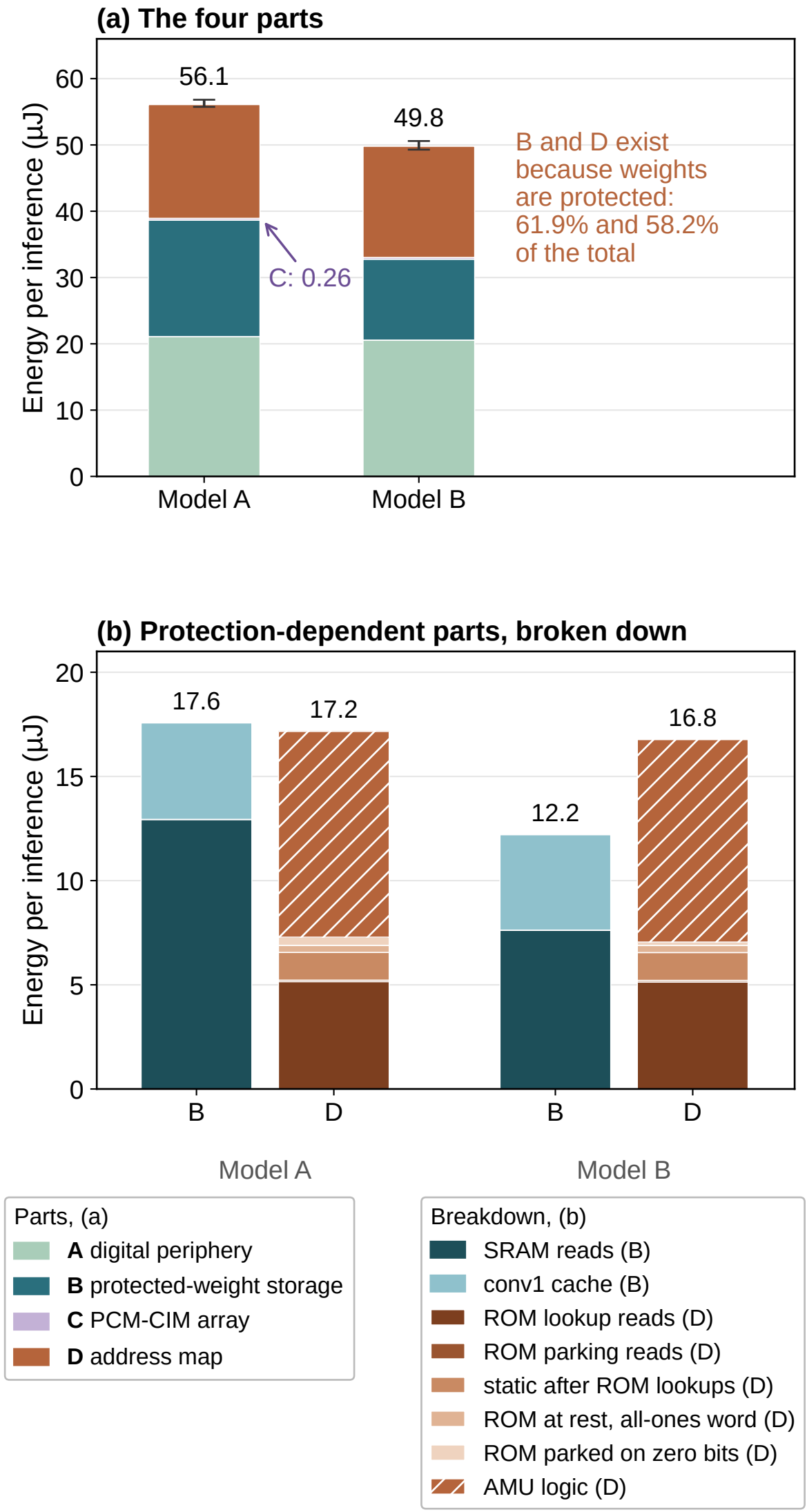
(a) The four parts
Energy per inference (µJ)
0
10
20
30
40
50
60
56.1
49.8
C: 0.26
B and D exist because weights are protected: 61.9% and 58.2% of the total
Model A
Model B
(b) Protection-dependent parts, broken down
Energy per inference (µJ)
0
5
10
15
20
17.6
17.2
12.2
16.8
B
D
B
D
Model A
Model B
Parts, (a)
A digital periphery
B protected-weight storage
C PCM-CIM array
D address map
Breakdown, (b)
SRAM reads (B)
conv1 cache (B)
ROM lookup reads (D)
ROM parking reads (D)
static after ROM lookups (D)
ROM at rest, all-ones word (D)
ROM parked on zero bits (D)
AMU logic (D)